# Blockchain

***A promising technology to positively alter nearly all the industries, including but not limited to the IT, Finance, Governance, Media, Medical and Law sectors.***

# Prerequisites for Blockchain Technology

**Understanding of cryptography:** Cryptography is the foundation of blockchain technology. A basic understanding of cryptographic concepts, such as hashing, public-key encryption, and digital signatures, is necessary.

**Distributed systems:** Blockchain is a distributed system that runs on multiple nodes. Therefore, it is essential to have a good understanding of distributed systems to build and deploy blockchain applications.

**Data structures and algorithms:** Blockchain technology relies on complex data structures such as Merkle trees and algorithms such as consensus algorithms. Understanding of these concepts is crucial for building a robust blockchain system.

# Prerequisites for Blockchain Technology

**Networking and security:** Blockchain technology requires a good understanding of networking protocols, such as TCP/IP, HTTP, and HTTPS. Additionally, a solid understanding of security concepts, such as firewalls, encryption, and authentication, is necessary to develop secure blockchain applications.

**Smart contracts:** Smart contracts are self-executing contracts with the terms of the agreement between buyer and seller directly written into lines of code. Knowledge of smart contract programming languages, such as Solidity, is necessary for building decentralized applications.

# Prerequisites for Blockchain Technology

**Business and economics:** Blockchain technology is disrupting traditional business models and creating new opportunities. Understanding the economics of blockchain and how it can be applied to business is essential for leveraging its potential.

**Legal and regulatory dimensions:** Blockchain technology operates in a regulatory gray area in many countries, and regulations are constantly evolving. Understanding the legal and regulatory environment in which blockchain operates is critical for creating compliant and successful blockchain applications.

# Blockchain Myths

## Blockchain is the same thing as Bitcoin (or any other cryptocurrency)

- There is a misleading idea that if you learn blockchain technology, you will become a good trader! This is untrue.
- Blockchain is not equivalent to any cryptocurrency, whether Bitcoin or any other currencies on the market like Altcoins.
- In fact, blockchain is a technology, whereas Bitcoin is a cryptocurrency that makes use of blockchain technology.
- Blockchain has many applications outside the crypto world.
- Blockchain technology provides a full support system for developing cryptocurrencies, whereas Bitcoin is a fundamental application that builds on this emerging blockchain technology.

# Blockchain can solve all security issues

- Blockchain cannot be used to definitively eliminate corruption or fraudulent activities.
- Blockchain's many applications have been developed by players in the various models governing economies around the world.
- Blockchain cannot all issues related to security. Solving all societal issues using blockchain is a formidable challenge. Thus, careful consideration needs to be given as to which societal issues should be addressed using blockchain.

# Blockchain is the only possible technology

- Blockchain is not necessarily the best technology for solving your problems;
- There might be better solved employing technology that does not use blockchain. It is possible that many different existing technologies would yield better results in terms of security without the use of blockchain.

# Blockchain and distributed databases are similar technologies

- Blockchain and distributed databases are different technologies.
- Blockchain is not a distributed databases.
- Blockchain is not designed to store and secure data.
- Blockchain and distributed databases are two different technologies, each with its own merits and demerits and different potential to solve different problems.
- Both are essential, and one cannot easily replace the other.

# History of Blockchain

A technology with the potential to become the foundation of global record-keeping systems, was introduced a mere decade ago by anonymous individuals associated with the digital currency Bitcoin, under the pseudonym Satoshi Nakamoto. Despite its relatively recent inception, blockchain has quickly gained recognition as a transformative innovation, poised to revolutionize various industries through its decentralized and secure nature.

# Milestones in Blockchain Development

**2008** – The publication of Bitcoin's whitepaper by Satoshi Nakamoto marked the groundbreaking introduction of the cryptocurrency. This event revolutionized the financial landscape, ushering in a new era of decentralized digital currency.

**2009** – The inaugural Bitcoin transaction between Satoshi Nakamoto and Hal Finney stands as a significant milestone in cryptocurrency history.

**2011** – Namecoin's launch marked a groundbreaking moment as it became the first alternative cryptocurrency to utilize blockchain technology. This pioneering step opened the door for a multitude of innovative blockchain-based digital assets.

# Milestones in Blockchain Development

**2013** – Vitalik Buterin's creation of Ethereum unleashed a revolutionary platform enabling the creation of smart contracts and decentralized applications (dApps). Ethereum's emergence introduced a new paradigm in blockchain technology, empowering developers to build complex applications on a decentralized network.

**2015** – The formation of the Enterprise Ethereum Alliance united leading corporations and blockchain startups, fostering collaboration in the advancement of blockchain technology.

**2016** – The Hyperledger Project, initiated by the Linux Foundation, set out to develop opensource blockchain software specifically tailored to enterprise applications.

# Milestones in Blockchain Development

**2017** – The cryptocurrency market witnessed an unprecedented surge in value, primarily led by Bitcoin, accompanied by an explosive growth in initial coin offerings (ICOs), fueling the development of innovative blockchain projects worldwide.

**2018** – Blockchain-based platforms like IBM's Food Trust have emerged as transformative solutions for supply chain management.

**2019** – Facebook's launch of the Libra cryptocurrency encountered substantial regulatory scrutiny and widespread resistance from governments worldwide.

**2020** – Major financial institutions have embraced blockchain technology. Many countries have launched their own central bank digital currencies (CBDCs), aiming to leverage the benefits of blockchain and enhance their monetary systems.

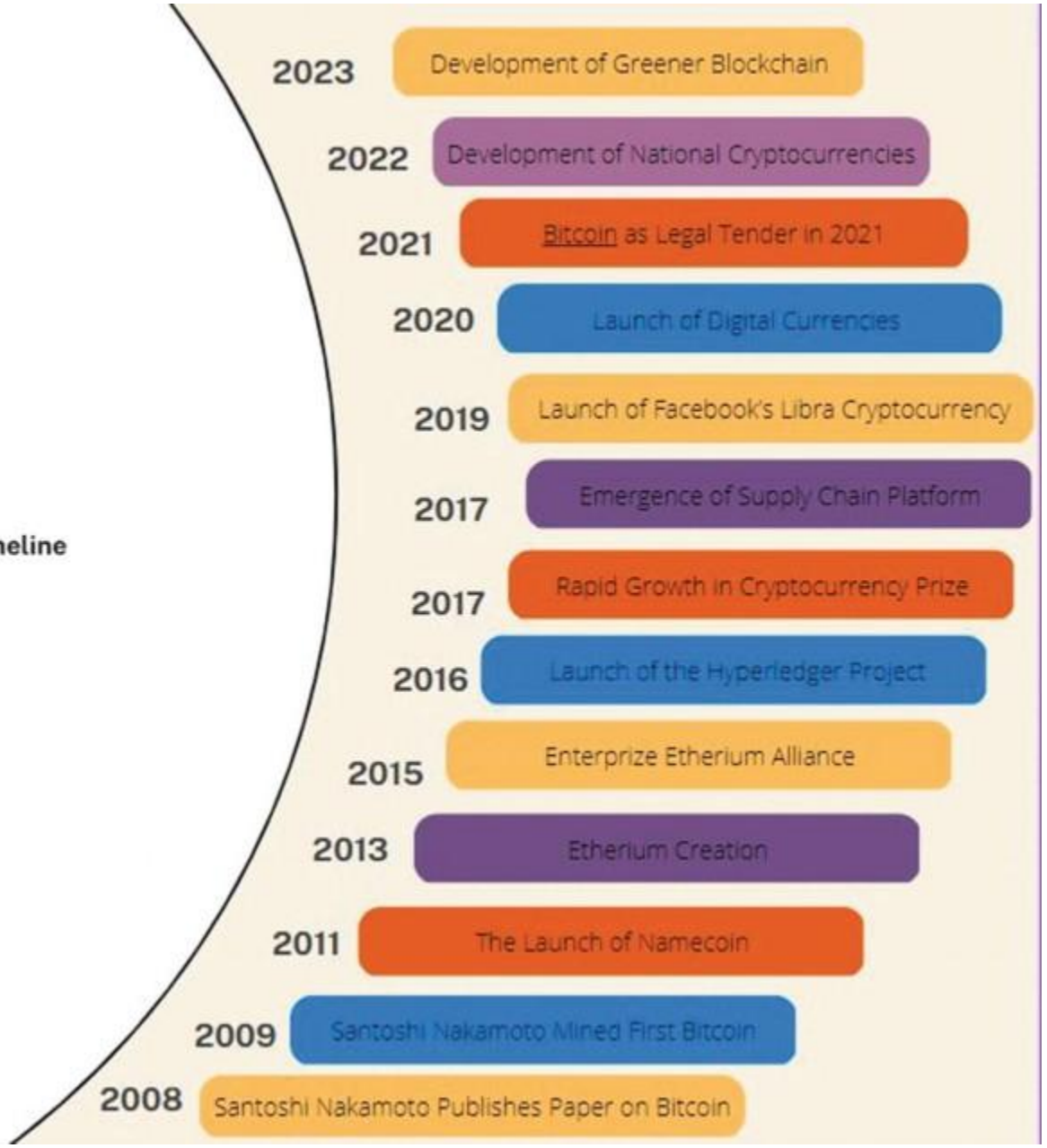
# Milestones in Blockchain Development

**2021** –El Salvador became the first country to officially adopt Bitcoin as legal tender in 2021. This decision enabled businesses to utilize Bitcoin for paying employee salaries and established its acceptance as a valid payment method throughout the country.

**2022** –The emergence of national cryptocurrencies revolved around the idea of CBDCs, where central banks opted to develop their own digital coins instead of relying on decentralized cryptocurrencies.

**2023** – The year 2023 has witnessed a notable focus on environmentally friendly blockchains, facilitated by carbon offsetting practices and energy-conscious network architectures.

# Blockchain Timeline



## Present Growth

- **Investment:** According to a report by CB Insights, global investment in blockchain startups has increased steadily, with over USD 8.2 billion raised across 342 deals in 2021 alone.
- **Enterprise adoption:** Major corporations, including IBM, Walmart, and Visa, are investing in and implementing blockchain technology for supply chain management, payment processing, and other applications.
- **Cryptocurrency adoption:** Cryptocurrencies such as Bitcoin and Ethereum have seen significant increases in adoption and investment. In 2021, the total market capitalization of all cryptocurrencies surpassed USD 2 trillion.

## Present Growth

**Government interest:** Several governments around the world are exploring the use of blockchain technology for various applications, including the development of CBDCs and voting systems.

**NFTs:** The emergence of nonfungible tokens (NFTs) on blockchain platforms has created a new market for digital assets and has the potential to revolutionize the art, music, and gaming industries.

**Increased scalability:** The development of new blockchain technologies, such as sharding and layer-2 solutions, is addressing the issue of scalability, making it possible to process more transactions per second and enabling more widespread adoption.

# Predicted Market

- The market for blockchain technology is expected to continue to grow in the coming years.
- The market size of blockchain technology globally was valued at USD 10.02 billion in 2022.
- It is projected to experience significant growth at a compound annual growth rate (CAGR) of 87.7% from 2023 to 2030.